Exp. No: 4.B

ENSEMBLE METHODS: RANDOM FOREST

Code:

```
# 1. Import Libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
  accuracy_score,
  precision_score,
  recall_score,
  f1_score,
  confusion_matrix,
  classification_report,
  roc_auc_score,
  roc_curve
)
# 2. Load Dataset
# Same dataset as Experiment 4A for comparison
data = load_breast_cancer()

X = pd.DataFrame(
  data.data,
  columns=data.feature_names
)
```

```
y = pd.Series(
  data.target,
  name="target"
)

# 0 = Malignant
# 1 = Benign

# 3. Train/Test Split (80/20)
# Stratification maintains the class distribution
X_train, X_test, y_train, y_test = train_test_split(
  X,
  y,
  test_size=0.20,
  random_state=42,
  stratify=y
)

# 4. Define Random Forest Model
# Random Forest does not require feature scaling
rf = RandomForestClassifier(
  random_state=42,
  n_jobs=-1
)

# 5. Define Hyperparameter Grid
param_grid = {
  "n_estimators": [100],
  "max_depth": [None, 10],
  "min_samples_split": [2],
  "min_samples_leaf": [1]
}
```

```
# 6. Hyperparameter Tuning using GridSearchCV
grid = GridSearchCV(
  estimator=rf,
  param_grid=param_grid,
  scoring="f1",
  cv=3,
  n_jobs=-1,
  verbose=0
)

# Train Grid Search
grid.fit(X_train, y_train)

# Display Best Parameters
print("Best Parameters from Grid Search:")
print(grid.best_params_)

# Get Best Random Forest Model
best_rf = grid.best_estimator_

# 7. Train Final Model
best_rf.fit(X_train, y_train)

# Predictions
y_pred = best_rf.predict(X_test)

# Probability predictions for ROC-AUC
y_prob = best_rf.predict_proba(X_test)[:, 1]

# 8. Calculate Evaluation Metrics
accuracy = accuracy_score(
  y_test,
  y_pred
)
```

```
precision = precision_score(
  y_test,
  y_pred,
  zero_division=0
)

recall = recall_score(
  y_test,
  y_pred,
  zero_division=0
)

f1 = f1_score(
  y_test,
  y_pred,
  zero_division=0
)

auc = roc_auc_score(
  y_test,
  y_prob
)

cm = confusion_matrix(
  y_test,
  y_pred
)

# 9. Display Results

print("\n========================================")
print("   RANDOM FOREST — TEST METRICS")
print("========================================")
```

```
print(f"Accuracy : {accuracy:.4f}")
print(f"Precision : {precision:.4f}")
print(f"Recall  : {recall:.4f}")
print(f"F1-Score : {f1:.4f}")
print(f"ROC-AUC : {auc:.4f}")

print("\nConfusion Matrix:")
print(cm)

print("\nClassification Report:")
print(
  classification_report(
    y_test,
    y_pred,
    target_names=["Malignant", "Benign"],
    zero_division=0
  )
)

# 10. Feature Importance — Top 10 Features

importances = pd.Series(
  best_rf.feature_importances_,
  index=X.columns
)

top10 = importances.sort_values(
  ascending=False
).head(10)

plt.figure(figsize=(8, 6))

top10[::-1].plot(
  kind="barh"
)
```

```
plt.xlabel("Importance")
plt.title("Top 10 Feature Importances — Random Forest")

plt.grid(
  axis="x",
  alpha=0.3
)

plt.tight_layout()
plt.show()

# 11. Plot ROC Curve

fpr, tpr, thresholds = roc_curve(
  y_test,
  y_prob
)

plt.figure(figsize=(7, 5))

plt.plot(
  fpr,
  tpr,
  label=f"Random Forest (AUC = {auc:.3f})"
)
plt.plot(
  [0, 1],
  [0, 1],
  linestyle="--",
  color="gray",
  label="Random Classifier"
)

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve — Random Forest")
```

```
plt.legend()
plt.grid(True)

plt.tight_layout()
plt.show()
```

Output:

```
Best Parameters from Grid Search:
{'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 100}

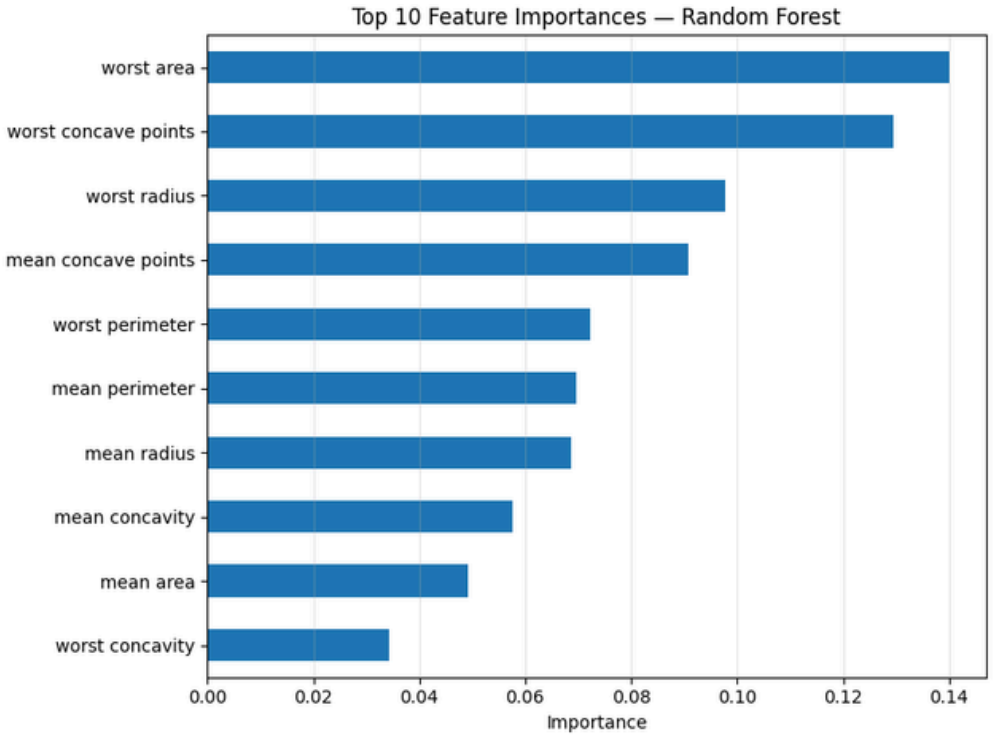
========================================
       RANDOM FOREST — TEST METRICS
========================================
Accuracy  : 0.9561
Precision : 0.9589
Recall    : 0.9722
F1-Score  : 0.9655
ROC-AUC   : 0.9937

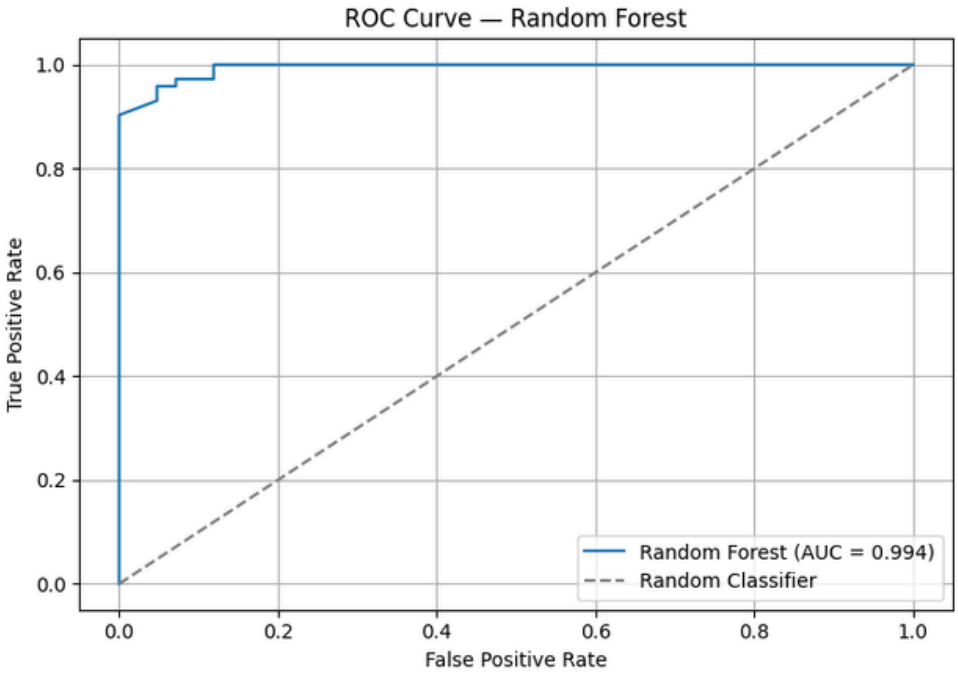
Confusion Matrix:
[[39  3]
 [ 2 70]]

Classification Report:
              precision    recall  f1-score   support

   Malignant       0.95      0.93      0.94        42
      Benign       0.96      0.97      0.97        72

    accuracy                           0.96       114
   macro avg       0.96      0.95      0.95       114
weighted avg       0.96      0.96      0.96       114
```




ROC Curve — Random Forest
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Random Forest (AUC = 0.994)
Random Classifier